1. Positional encoding
2. Self Attention → Multihead Self Attention
3. Layer normalization

Decoder

- Cross Attention
- Self mask attention

end2end flow encoder → decoder

2-3

API → llm

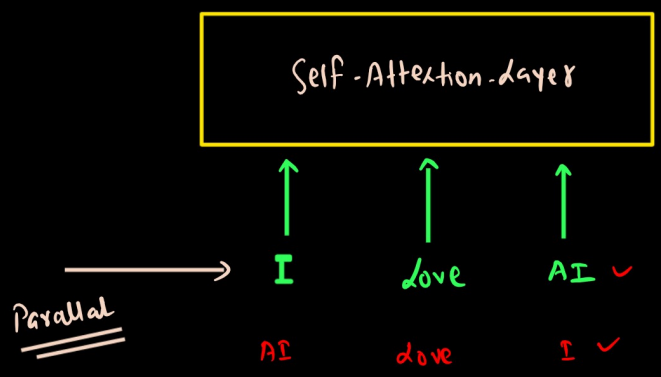
FT → llm

hf, unsloth

coding

Positional encoding

Problem statement → Data → Parallelly Solⁿ → Sequence → PE



Positional Encoding (Definition)

Positional encoding is a technique used to inject information about the position (order) of tokens into their embeddings, so that the model can understand sequence order.

Why it is needed?

Transformers process all tokens in parallel

So they don't know order by default

Example:

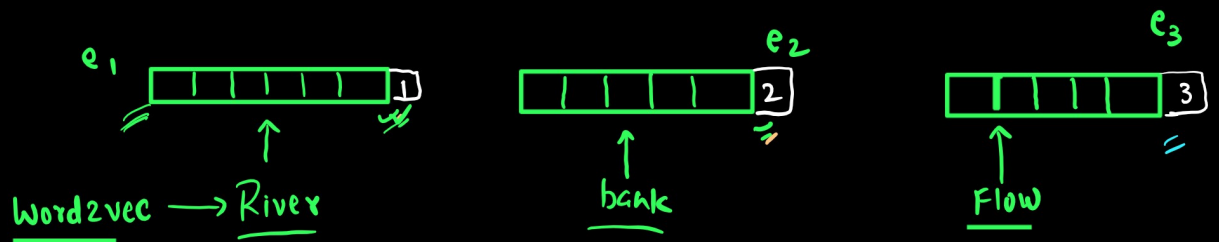
- "I love AI"
- "AI love I"

Without positional encoding → both look same to model

Dog bites man

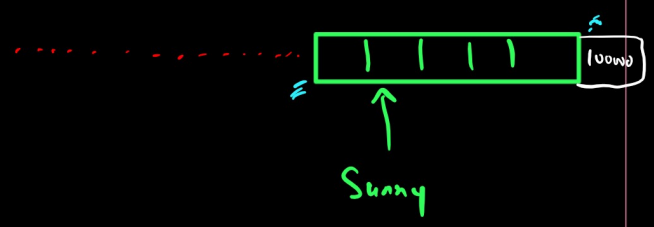
man bites dog

Self - Attention - Layer



Real time data → Self → 100000

No-limit



Simple - Approach → Self → 4 token

I	Love	AI	Much
1	2	3	4

Self → 400 token

I	Love	AI	Much	Again	.....	You
1	2	3	4	5	.....	400

Self → Limitly

n token

1L, 10L ...

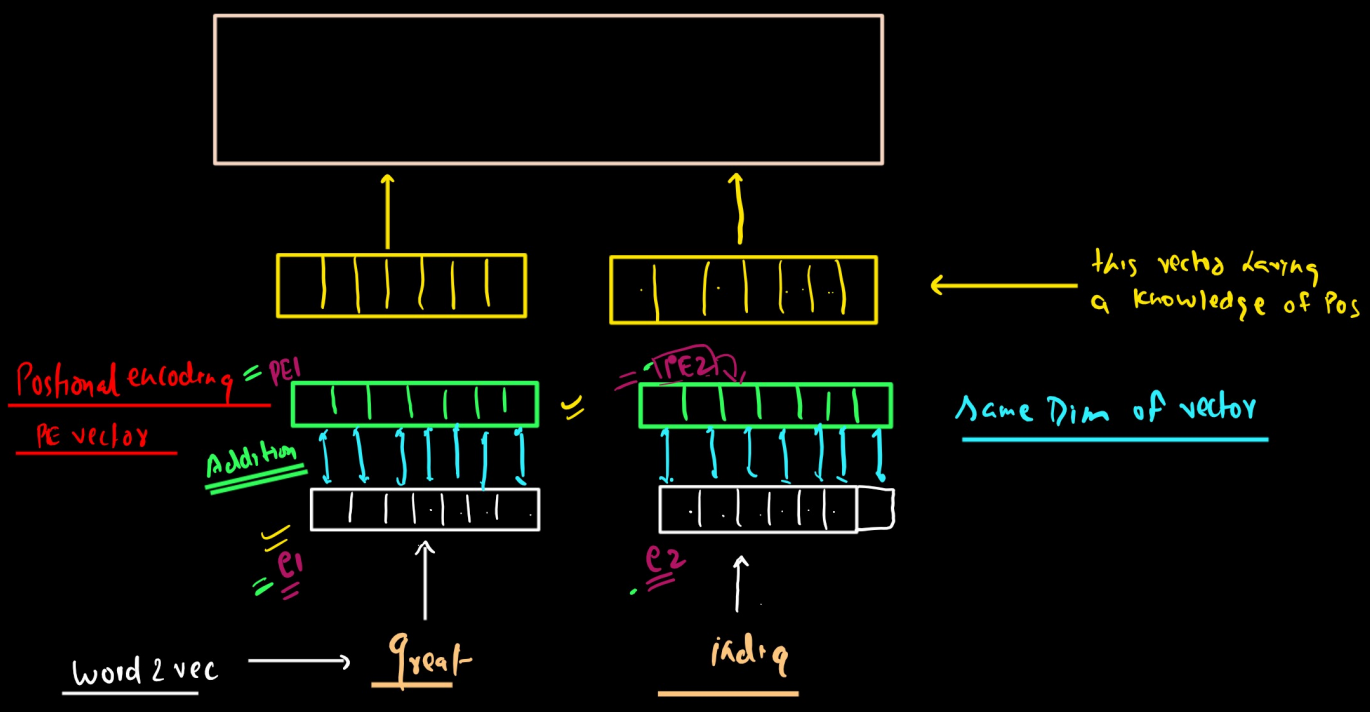
Drawback

1. no of token could be limited
2. Discreate (1, 2, 3, 4, 5 ... 100, 10000) → vector

↓

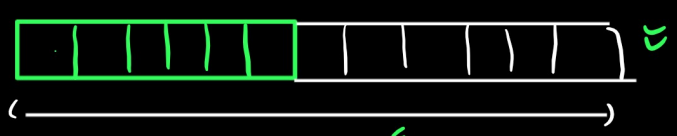
NN

(training might become unstable)



Concatination

(APPEND) Dim↑↑↑↑



10-Dim

NN / Attention

more computational Power

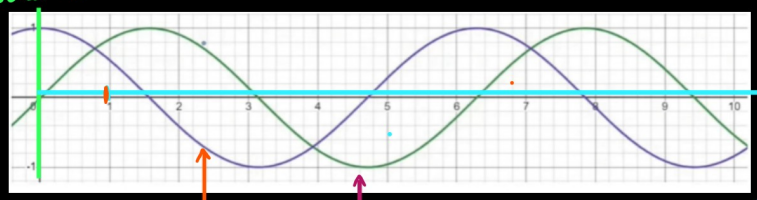
Addition

$e_1 + PE1$

same dim of vector

PE1

Encoded value



Position of the word

cos [-1 to +1]

Sin → [-1 to +1]

PE2

9-10th class

School

Pythagoras

TIGNOMETRY

Sin, cos, tan

Sin | cos

Original RP 512

= Word2vec

Great

1

India

2

~~Descrete~~

Continuous

E1

6

Sin(Pos)

PE1 → Sin(1) → 0.84

Cos(1) → 0.54

Cos (Pos)

| 0.84 | 0.54 |

2

E2

6

Sin (Pos)

Sin(2) → 0.9

Cos(2) → -0.4

Cos (Pos)

| 0.9 | -0.4 |

2

PE2

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

$\sin(Pos) \longrightarrow \sin(Pos / 10000^{2i/d_{model}})$

$\cos(Pos) \longrightarrow \cos(Pos / 10000^{2i/d_{model}})$

Pos $\longrightarrow$ Position

dmodel $\longrightarrow$ Dim of embedding model

i $\longrightarrow$ Range

$i = [\text{Low Range} - \text{Upper Range}]$

$= \left[0 - \left(\frac{d_{model}}{2}\right) - 1\right]$

$i = \left[0 - \left(\frac{6}{2} - 1\right)\right]$

$i = [0 - 2] \longrightarrow \{0, 1, 2\}$

6

(wordzvec model)

great

rading

4 tokens

	I	Love	my	country
Pos $\longrightarrow$	0	1	2	3

Dim → 6

Pos = 0
great

E1

Pos = 1
india

E2

word2vec

$d_{model} \longrightarrow 6$

$i = [0 - (6/2 - 1)]$

$= [0 - 2]$

$i = 0, 1, 2$ → sin/cos, cos/sin, sin/cos → 6

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{model}})$$
$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{model}})$$

PE1

PE2

For i = 0 → Pos 0

$PE(0,0) = \sin(0/10000^0) = 0$

$PE(0,1) = \cos(0/10000^0) = 1$

For i = 0 → Pos → 1

$PE(1,0) = \sin(1/10000^0) = 0.84$

$PE(1,1) = \cos(1/10000^0) = 0.54$

For i = 1 2×1

$PE(0,2) = \sin(0/10000^{1/3}) = 0$

$PE(0,3) = \cos(0/10000^{1/3}) = 1$

2×1+1

For i = 1

$PE(1,2) = \sin(1/10000^{1/3}) = 0.04$

$PE(1,3) = \cos(1/10000^{1/3}) = 0.99$

For i = 2

$PE(0,4) = \sin(0/10000^{2/3}) = 0$

$PE(0,5) = \sin(0/10000^{2/3}) = 1$

For i = 2

$PE(1,4) = \sin(1/10000^{2/3}) = 0.01$

$PE(1,5) = \cos(1/10000^{2/3}) = 0.99$

PE1 | 0 | 1 | 0 | 1 | 0 | 1 |

+

Dim 6

E1 → x_1

PE2 → | 0.84 | 0.54 | 0.04 | 0.99 | 0.01 | 0.98 |

+

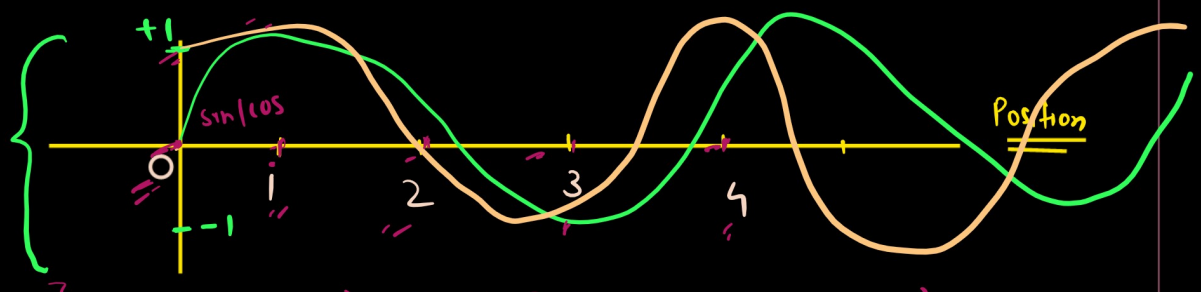
E2 → x_2

→ TODEY I AM TEACHING PE

token → 5

Pos → 0 1 2 3 4

~~descrete~~ Continuous (sin/cos)



encoded value

seq info

[. . . . .]

② Self Attention ⟶ Multihead Attention

Money [bank] grow

Similarity ⟶ Dot Product

$$bank = w_1 * Money + w_2 * bank + w_3 * grow$$

(q,k)

$$[\underbrace{e_{bank}}_{q} \cdot \underbrace{e_{money}}_{k}]\, Money + [\underbrace{e_{bank}}_{q} \cdot \underbrace{e_{bank}}_{k}] * bank + [\underbrace{e_{bank}}_{q} \cdot \underbrace{e_{grow}}_{k}] * grow$$

$$bank = \underbrace{(e_{bank} * e_{money}) * e_{money}}_{S_1} + \underbrace{(e_{bank} * e_{bank}) * e_{bank}}_{S_2} + \underbrace{(e_{bank} * e_{grow}) * e_{grow}}_{S_3}$$

SoftMAX

$$w_1 = \frac{e^{S_1}}{e^{S_1}+e^{S_2}+e^{S_3}} \qquad w_2 = \frac{e^{S_2}}{e^{S_1}+e^{S_2}+e^{S_3}} \qquad w_3 = \frac{e^{S_3}}{e^{S_1}+e^{S_2}+e^{S_3}}$$

value vector (v)

$$w_1 * e_{money} + w_2 * e_{bank} + w_3 * e_{grow}$$

(v) (v)

{Self → (Q,K,V)}

bank ⟵ Contextual vector

| 1 | 2 |

| 3 | 4 |

1	2
a_1	b_1

3	a_2
4	b_2

$$\underbrace{(1 \times 3)}_{a_1 * a_2} + \underbrace{(2 \times 4)}_{b_1 \times b_2} = 11$$

Scalar value

Self-Attention $\longrightarrow$ Q, K, V

$$D = \{a:1,\ b:2,\ c:3\}$$

Analogy

$D[a] \longrightarrow$ o/p $= 1$

query

key

value

Self-Attention $\rightarrow$ extend

m self-Attention

Cross Attention

masked " "